AGENTHON 2026
TRACK 2 QUANT V1

Quant v1 Development Report

From the Gaussian baseline to regime-aware bootstrap experiments
Model structure, validation design, and ablation results from a QRM perspective

Report date: 2026-09-28

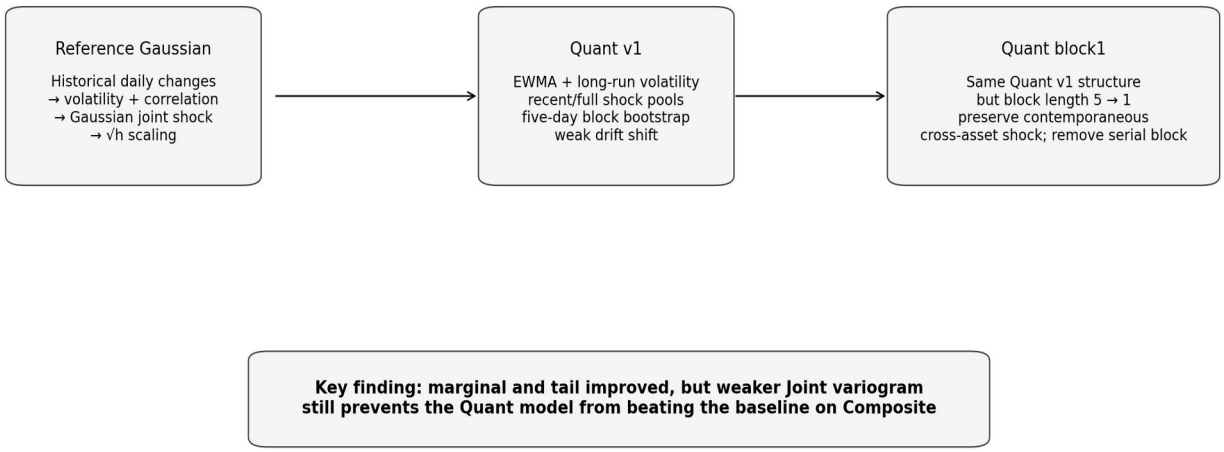
Audience: QRM readers more familiar with probability, statistics, and risk modelling than Python implementation

1. One-page summary

The core of the work so far was to move beyond the existing Gaussian random-walk baseline, build a Quant model that better reflects recent volatility regimes and fat tails, and compare it with the baseline through walk-forward tests at multiple historical dates.

The current conclusion is clear. The new Quant model improved the individual predictive distributions (Marginal CRPS) and tail calibration relative to the baseline, but lost ground on the Joint variogram that evaluates co-movement across assets. As a result, it still did not beat the baseline on the final Composite score.

Item	Current status
Official submission format	admissible = true; gates g0-g3 all pass
Baseline	Reference Gaussian random walk
Quant v1	EWMA volatility + recent/full mixture + multivariate block bootstrap + weak drift
Validation	24-fold walk-forward, spacing 21, 1,000 Monte Carlo draws
Current best Quant	quant_block1 (block length 5 → 1)
Main gain	Marginal -3.2%, Tail -19.7% vs reference
Main problem	Joint +5.1% worse → Composite +2.2% higher (worse) than reference
Next direction	Quant v2: retain strong marginal/tail performance while stabilizing correlation/dependence



Rather than explaining Python line by line, this report focuses on the statistical model implemented by each component and on why the empirical results behaved as they did.

2. Forecasting problem and scoring function

Track 2 is not a point-forecasting problem. Participants submit Monte Carlo draws across multiple assets and horizons, and the realized outcome is used to evaluate how well the full predictive distribution performed. A strong model therefore needs to capture uncertainty, dependence, and tails rather than merely getting the mean right.

$$\mathbf{S = 0.5 \times Marginal\ CRPS + 0.3 \times Joint\ Variogram + 0.2 \times Tail\ Penalty}$$

Lower is better for every component. Because Joint carries a 30% weight, dependence errors are expensive.

2.1 Marginal CRPS - Is each asset distribution calibrated?

$$\mathbf{CRPS(F,y) = E|X-y| - \tfrac{1}{2}\, E|X-X'|}$$

Intuitively, CRPS asks whether the forecast distribution is appropriately concentrated around the realized value y. A distribution that is too narrow is penalized when it misses the realization, while a distribution that is too wide is also penalized. From a QRM perspective, it is a proper scoring rule that evaluates calibration and sharpness of the entire predictive distribution rather than a single VaR point.

2.2 Joint variogram - Is cross-asset co-movement realistic?

$$\mathbf{VS \approx \Sigma_{(i<j)}\, w_{ij}\, [\ |y_i-y_j|^p - E|X_i-X_j|^p\ ]^2}$$

Even if UST 2Y, 5Y, 10Y, and 30Y are each forecast well marginally, the Joint score deteriorates when a single draw produces unrealistic yield-curve twists or incorrect co-movement across tenors. This emerged as the main weakness of the current Quant model.

2.3 Tail penalty - Are extreme quantiles placed correctly?

Pinball loss is evaluated at the 1%, 5%, 95%, and 99% quantiles. Models that better capture the current volatility regime and fat tails can therefore benefit. In the current experiments, the Quant model improved clearly over the baseline on this component.

3. Starting point: Reference Gaussian baseline

The reference model uses historical daily changes to estimate per-asset scale and cross-asset correlation, then generates joint Gaussian shocks for the h-day predictive distribution. For level targets, it is close to a classical random-walk approximation centered on the current level, with uncertainty increasing approximately with the square root of h.

$$X_{t+h} \approx X_t + \sqrt{h} \cdot D \cdot z, \qquad z \sim N(0,R)$$

D = diagonal matrix of daily asset volatilities; R = historical empirical correlation

The main strengths of this structure are simplicity and stability. For a UST curve with strong correlation across tenors, the empirical-correlation plus Cholesky structure preserves joint dependence relatively well.

The weaknesses are regime sensitivity and non-normality. If volatility has risen recently, using a variance estimate dominated by the full twenty-year history may be too slow to adapt; treating financial fat-tailed shocks as Gaussian can also weaken tail calibration. The Quant candidate was built to address these issues.

Baseline strengths	Baseline limitations
Simple and reproducible	May react too slowly to recent volatility regimes
Stable cross-asset correlation structure	Gaussian tails may be too thin for financial data
Clear horizon interpretation via √h scaling	Uses little nonlinear or mean-reversion signal

4. What changed in Quant v1?

Quant v1 is not a complex machine-learning model that discards the baseline. It is a conservative probabilistic forecasting engine: volatility reacts more strongly to recent information, the shape of shocks is resampled from actual historical innovations, and drift is allowed only weakly.

Parameter	Current value	Meaning
recent_window	126	Business days used in the recent shock pool
recent_mix	0.60	60% probability of sampling a bootstrap block from the recent pool
ewma_half_life	60	EWMA half-life; not “use only the latest 60 days”
ewma_weight	0.70	70% EWMA covariance + 30% long-run covariance
block_length	5	Sample historical shocks in consecutive five-day blocks
numeric_drift_shrink	0.20	Use only 20% of the estimated drift
max_drift_sigma	0.35	Cap drift magnitude at 0.35 × horizon σ
min_pool_rows	30	Minimum aligned history required

4.1 EWMA volatility: half-life 60, not a 60-day window

$$\lambda = \exp[\ \ln(0.5) / 60\ ], \qquad \text{weight(age)} \propto \lambda^{\text{age}}$$

An observation 60 business days old receives roughly half the relative weight of today's observation, while one 120 business days old receives roughly one quarter. The data are not truncated after 60 days; instead, more recent observations receive larger weights.

$$\Sigma_\text{blend} = 0.70\ \Sigma_\text{EWMA} + 0.30\ \Sigma_\text{full}$$

Each asset's target daily volatility is then taken from the diagonal of the blended covariance matrix: $\sigma_i = \sqrt{\Sigma_{ii}}$. Importantly, the current implementation does not directly impose the blended covariance off-diagonal terms on simulation dependence. This becomes a major improvement target for the next version.

5. Understanding Quant v1 bootstrap and drift

5.1 Multivariate block bootstrap

Historical shocks are not sampled independently by asset. The vector of shocks across assets on the same date is treated as one row and sampled jointly, preserving contemporaneous cross-asset structure to some extent.

$$\varepsilon_t = (\varepsilon_{1t}, \varepsilon_{2t}, \ldots, \varepsilon_{kt}) \rightarrow \text{sample the entire row jointly}$$

In v1, each selection takes a five-day consecutive block. The intention was to preserve short-run serial structure. However, the 24-fold ablation improved when block length was reduced to 1, suggesting that the five-day serial block added noise or excessive path dependence in this task.

5.2 Recent pool and full-history pool

Two standardized shock pools are constructed: one from the full history and one from the most recent 126 business days. For each draw block, the recent pool is selected with 60% probability and the full-history pool with 40% probability. This is a compromise between emphasizing the current regime and avoiding over-reliance on only the latest six months.

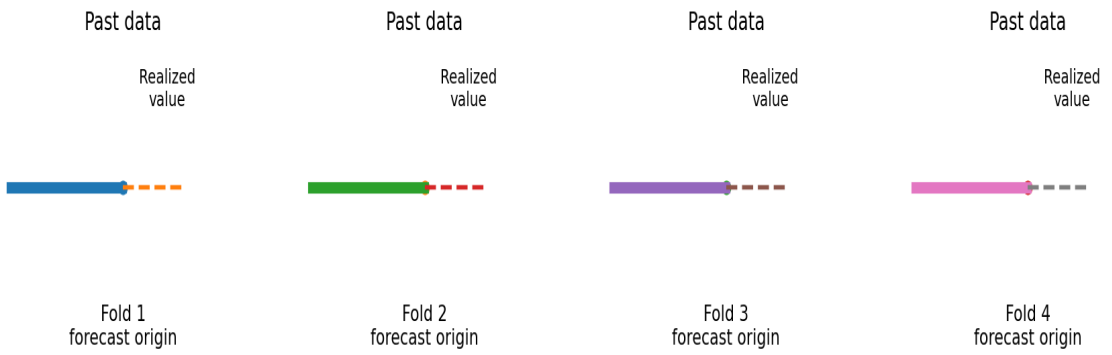
5.3 Weak mean-reversion / trend shift

$$\text{Future_h} = \beta_0 + \beta_1 \cdot \text{PriorRunup_h} + \text{error}$$

$$\text{Applied shift} = 0.20 \times \text{fitted value}, \ \text{capped at } \pm 0.35 \times \sigma_h$$

For level targets, a simple regression links the previous h-day run-up to the subsequent h-day change, but only 20% of the fitted value is used as a center shift. In QRM terms, the signal is heavily shrunk and capped rather than bet on aggressively. Because removing drift had only a small effect in the ablation, the main current bottleneck is dependence rather than drift.

6. Walk-forward validation: testing by going back in time



One fold = forecast using only information available at a historical origin, then compare with the subsequent realization.

One fold chooses a historical date as the as-of date, builds a forecast distribution without using information after that date, and then compares it with the subsequent realization. Repeating this pseudo-out-of-sample test across many origins reduces the risk of mistaking a model that happened to work in one month for a genuinely strong model.

The final comparison used the long UST history in the t2-F3-election-2024-joint unit, with 24 folds, 21 business rows of spacing between folds, and 1,000 Monte Carlo draws per model per fold.

Validation setting	Value
Unit	t2-F3-election-2024-joint
Folds	24
Spacing	21 business rows
Monte Carlo draws	1,000 per model per fold
Number of models	7 (reference + 6 Quant variants)
Scoring	Marginal / Joint / Tail / Composite

Technical issues encountered during validation

- The official scorer confirmed quant_out/forecast.parquet as admissible=true with gates g0-g3 passing.
- Public units do not contain sealed realized outcomes, so the official local scorer performs gate validation rather than computing a true leaderboard accuracy score.
- On Windows, qfbench2-smoke failed because O_NOFOLLOW / O_DIRECTORY are unavailable. This is a platform filesystem limitation rather than a model-logic failure.
- The initial exemplar walk-forward failed because the harness panel-path assumption did not match the exemplar file layout. Research walk-forward tests were subsequently run on a long-history practice unit.

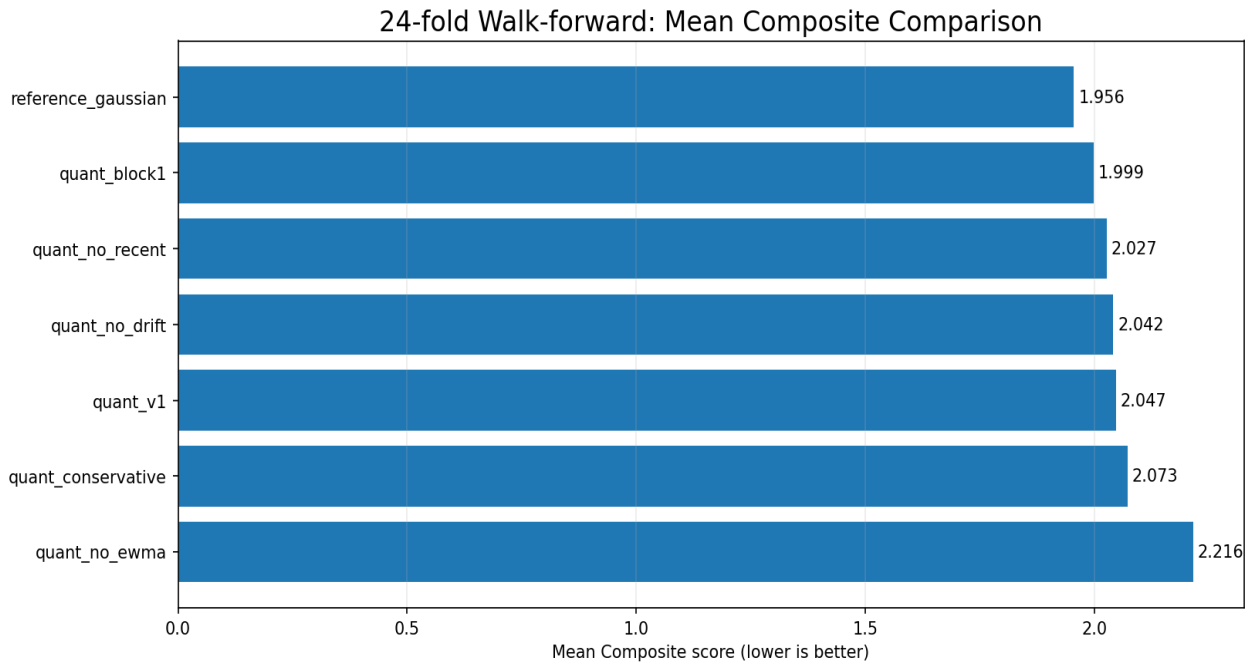
7. Ablation study: which components actually helped?

An ablation study removes or weakens one component at a time to test whether that component actually contributes to performance. From a QRM perspective, the logic is similar to sensitivity analysis or nested-model comparison.

Model	Change vs v1	Question
reference_gaussian	Existing Gaussian baseline	Overall benchmark
quant_v1	Original Quant configuration	Current candidate
quant_no_recent	recent_mix = 0	Is the recent shock pool useful?
quant_no_ewma	ewma_weight = 0	Is recent-volatility weighting useful?
quant_no_drift	drift shrink = 0	Is the mean-reversion/trend center shift useful?
quant_block1	block_length = 1	Does the five-day serial block actually help?
quant_conservative	recent=.25, ewma=.40, drift=.10	Does a more conservative setting improve results?

The objective was not merely to tune parameters for a better score, but to identify which modelling structures were useful and why.

8. 24-fold results



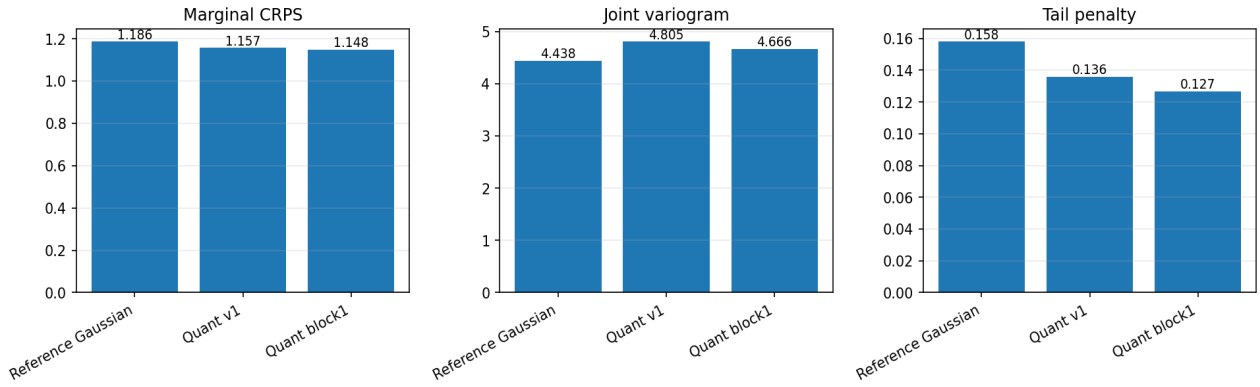
Model	Marginal ↓	Joint ↓	Tail ↓	Composite ↓	Win vs Ref
reference_gaussian	1.1858	4.4381	0.1579	1.9559	-
quant_block1	1.1477	4.6657	0.1268	1.9989	41.7%
quant_no_recent	1.1561	4.7427	0.1316	2.0272	41.7%
quant_no_drift	1.1548	4.7912	0.1349	2.0418	37.5%
quant_v1	1.1568	4.8047	0.1358	2.0470	29.2%
quant_conservative	1.1672	4.8701	0.1411	2.0728	41.7%
quant_no_ewma	1.2046	5.2636	0.1736	2.2161	33.3%

Reference Gaussian achieved the lowest mean Composite at 1.9559. Among the Quant variants, quant_block1 was best at 1.9989. The current best Quant therefore remained about 2.2% higher, and thus worse, than the baseline on Composite.

The component scores show that this was not a simple failure. Relative to the reference, quant_block1 improved Marginal by about 3.2% and Tail by about 19.7%, but worsened Joint by about 5.1%. Because Joint carries a 30% weight, the dependence loss offset the gains in Marginal and Tail.

9. Interpreting the results from a QRM perspective

Where the model improved - and where it lost ground



9.1 EWMA volatility was useful

quant_no_ewma produced a Composite of 2.2161, about 8.3% worse than v1 at 2.0470. Removing recent volatility information worsened marginal, joint, and tail scores. The core idea of giving greater weight to the recent volatility regime is therefore worth retaining.

9.2 The five-day block was excessive

Reducing block length from 5 to 1 improved Composite by about 2.35% relative to v1. Because the contemporaneous multivariate shock vector was retained while only the serial block was removed, the five-day path dependence appears to have contributed more noise than information in this task.

9.3 The recent pool and drift were secondary issues

Removing recent_mix entirely improved on v1 by roughly 1.0%, while removing drift improved it by about 0.25%. The direction is informative, but the effect is weaker than the block-length result. The priority should therefore be stabilizing joint dependence rather than making drift more elaborate.

9.4 The key bottleneck is not covariance itself, but the dependence structure actually used

The code computes a covariance blend of Σ_{EWMA} and Σ_{full}, but the final bootstrap scale primarily uses the per-asset σ values from the diagonal. Cross-asset dependence is determined indirectly by the standardized historical row bootstrap. In other words, the volatility model improved, but the correlation model still depends on bootstrap sample selection without separate control. This is a plausible source of the Joint variogram loss.

10. Translating the code changes into statistical modelling terms

Code/model change	Statistical meaning	Result
EWMA half-life 60	Give more weight to recent shocks	Keep it; removing it materially worsened performance
EWMA 70% + Full 30%	Blend short-run regime sensitivity with long-run stability	Key candidate behind current marginal/tail improvement
Recent window 126 / mix 60%	Reuse recent six-month shocks more often	Limited effect; may be too aggressive
Block length 5	Preserve five-day shock clusters	Disadvantageous in this UST test
Block length 1	Preserve contemporaneous cross-asset shocks only	Current best Quant setting
Drift shrink 20% + cap	Weak mean-reversion/trend signal	Small impact
Ablation harness	Compare multiple settings on identical folds	Identifies improvement directions from data

The important point is not that a large amount of Python code was changed. The modelling assumptions were changed one at a time and their economic and statistical implications were tested. In QRM terms, the exercise separately examined the volatility estimator, dependence estimator, innovation distribution, and conditional mean.

$$\text{Forecast distribution} = \text{Conditional mean} + \text{Volatility scale} \times \text{Joint shock distribution}$$

The evidence so far suggests that the volatility scale improved more clearly than the conditional mean, while the joint shock distribution and dependence remain less stable than in the baseline.

11. Next step: design direction for Quant v2

Rather than adding a new layer of complexity, the next version should combine what the current Quant model does well with what the baseline does well. The key idea is to separate volatility from correlation.

$$\Sigma_t = D_t R_t D_t$$

D_t represents EWMA-based volatility that responds quickly to the recent regime, while R_t is a more stable correlation matrix estimated from a longer sample with shrinkage. This is the familiar QRM decomposition of covariance into volatility × correlation × volatility.

Quant v2 component	Plan
Volatility D_t	Keep EWMA + long-run blend
Correlation R_t	Blend/shrink long-run and recent/EWMA correlation separately
PSD stabilization	Keep eigenvalue floor / nearest-PSD repair
Innovation	Fat-tail historical bootstrap or recolored bootstrap
Block	Start with length = 1
Drift	Start at zero or with very small shrinkage
Validation	Compare reference / block1 / v2 on the same 24 folds

Success condition

The first objective is to preserve roughly the current best Quant levels of Marginal 1.1477 and Tail 0.1268 while reducing Joint from 4.6657 toward the reference value of 4.4381. Achieving this would create room to overtake the baseline on Composite.

Caveats

- The current 24-fold result is a research diagnostic from a single UST cross-asset unit and should not be treated as representative of all Track 2 tasks.
- The next stage should test whether the same structure is stable not only for rates, but also for FX, factor returns, and other event periods.
- Overfitting model selection to the public historical backtest is a risk, so the number of parameters should remain limited and the model structure should stay economically interpretable.

12. Ten points to remember at this stage

1) The submission format is valid: official admissibility gates g0-g3 passed.

2) The value 60 is an EWMA half-life, not a rule that only the most recent 60 days are used.

3) Quant v1 added EWMA volatility, a recent/full shock mixture, block bootstrap, and a weak drift term.

4) One walk-forward fold is a single pseudo-out-of-sample test at a historical origin.

5) The reference and multiple Quant variants were compared under the same conditions across 24 folds.

6) Removing EWMA materially worsened performance. Recent volatility-regime information is useful.

7) Reducing the five-day block to one day produced the best Quant performance.

8) The best Quant variant outperformed the reference on Marginal and Tail.

9) However, the weaker Joint variogram means the reference still wins on final Composite.

10) The next objective is Quant v2: combine strong volatility/tail modelling with stable correlation.

Reference command used for the current experiment

```
python tools\walk_forward_quant.py `
  --unit units\t2-F3-election-2024-joint `
  --folds 24 `
  --spacing 21 `
  --n-draws 1000 `
  --out research\ust_quant_ablation_24fold.csv
```

This report is based on the candidate repository code, official local gate output, and walk-forward/ablation results verified in the current work. Public units do not contain sealed leaderboard realized outcomes; the figures here are research pseudo-out-of-sample diagnostics, not official final competition scores.